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Near-offset Trace Reconstruction from Sparse Ocean Bottom Seismometer Data Using Seismic Interferometry

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SUMMARY

Ocean bottom seismometer data are often acquired using large receiver intervals; for example large cable spacing in cross-line ocean bottom cable data acquisition. In this case, the missing of near-offset traces may result in poor illumination of the shallow subsurface by the primary reflections. However, the surface-related multiple reflections have a better illumination of this region even if the receiver array is sparse. Using a combination of the seismic interferometry and the data-driven FX-interpolation methods, we extrapolated the surface-related multiple wavefield of a dense array of shots and a sparse array of receivers towards a primary wavefield corresponding to dense arrays of both, shots and receivers. We applied our method to a synthetic dataset to reconstruct the near-offset traces and showed that our method provided similar seismic image of shallow structures compared with densely acquired seismic data.

Introduction

Ocean-bottom seismometer (OBS) surveys have advantages over towed streamer surveys, especially in shallow water environments and transition zone areas. However, the spatial sampling of the shallow reflections is poor if the spacing of the OBS is sparse. In such a situation, the traces at near offsets play an important role to image the near subsurface.

Through cross-correlations, the seismic interferometry (SI) can redatum the recorded wavefields into different survey configurations (Schuster *et al*. 2004). The redatumed wavefields can be treated as if the reconstructed traces were produced by virtual shots located at the real receiver position. This technique is called the virtual shot method (Bakulin and Calvert 2006; Wapenaar *et al*. 2011; Schuster *et al*. 2004). For OBS, the SI can convert every source into a virtual receiver if the receiver spacing is fine, and the surface-related multiples (SRM) are taken into account (Hanafy and Schuster 2014). However, artefacts may appear in the reconstructed wavefield if the receiver spacing is very large.

We propose here a method for retrieving a dense streamer-like acquisition using a dense source spacing (e.g., of 20 m) and a sparse OBS spacing (e.g., 320 m). Our technique is based on a combination of the SI and data-driven FX-interpolation to remove artefacts caused by the sparseness. We show that near-offset traces can be accurately retrieved using the SRM and they can provide seismic imaging that can be compared with conventional imaging using densely acquired dataset.

Method

Shallow reflectors are not well sampled by primary reflections if an OBS array has a large receiver interval. Figure 1a shows schematic ray paths for primary reflections recorded by receivers at position R_k and generated by a source at position S_i. The red portions indicate the part of the reflector that is illuminated by primary reflections. By considering the SRM, we can finely sample this reflector (Figure 1b, green). However, it is preferred to remove SRM and work only with primary reflections for conventional seismic processing and imaging.

The SI by cross-correlations can redatum SRM towards primary reflections (Hanafy and Schuster 2014). Figure 1b shows the ray path of a multiple recorded by the receiver at R_k and produced by the source at S_i. A part of this ray path coincides with the ray path of a multiple at the sea floor generated by source at S_j and recorded by the same receiver at R_k. The cross-correlation of these two traces can be seen as removing this common part of those ray paths. Therefore, the result is a virtual trace with a primary reflection recorded at S_j and produced at S_i. Taking into account two adjacent receivers with a large spacing between them, we can obtain two virtual traces with primaries that finely sample the shallow reflector (Figure 1c).

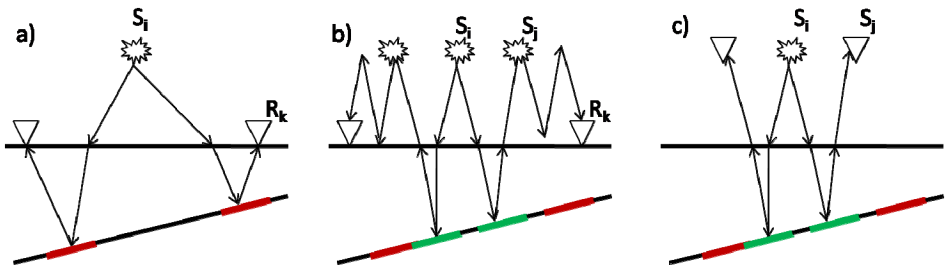


Figure 1 *(a) Poor spatial sampling (in red) of the primaries when ocean bottom receiver spacing is large. (b) The SRM can be treated as primaries by cross-correlating traces recorded by receiver at position R_k and produced by sources at positions S_i and S_j. (c) Primaries with finer spatial sampling (in green) obtained after applying seismic interferometry.*

Theoretically, the SI requires a surface integral of an infinite number of receivers. However, the SI summation can be applied on traditional seismic acquisitions that have a limited number of receivers if the wavefield is recorded in the stationary phase region. The stationary phase region corresponds to

the zone where the summation has the main contribution (Snieder 2004). In the case of Figure 1b, the receiver R_k is located at the stationary phase region to redatum the multiple towards a primary after the SI summation. The sparseness of the receivers also affects the SI summation and may introduce artefacts. Also, we can reduce the non-physical artefacts of the SI following the method proposed by Hanafy and Schuster (2014): For one input, we use the synthetic dataset generated only for the seafloor reflections. In real cases, a smooth seafloor model can be used. For the second input, we used a multiple model of the sub-surface that can be generated with SRM modelling (e.g., Pica *et al*. 2005).

In order to solve the sparseness, we propose generating the missing traces on the cross-correlograms by using the FX-interpolation (Spitz 1991). However, implementing the FX-interpolation on the cross-correlograms is not straightforward. We propose two adjustment operations to flatten the wavelets of traces before the FX-interpolation and to avoid spatial aliasing. One consists in a quadratic adjustment in which we choose a parameter that flattens the parabolic shape due to the spreading of the wavefield recorded at the sea floor. The second one is a linear parameter that flattens a dipping reflector event. We can obtain an approximation of these two flattening parameters by analysing the parabolic and dipping shape on the cross-correlogram while working at low frequencies. We prefer to set these flattening parameters on the cross-correlograms because it allows more degrees of freedom compared to setting them for common shot or common receiver gathers.

Figure 2a shows the initial cross-correlogram for a large receiver interval. In Figure 2b, the trace reconstructed after stacking the cross-correlogram (in green) shows a large amount of artefacts compared to the trace reconstructed using finer receiver interval (in blue). Then, we apply the flattening operations and the FX-interpolation to the cross-correlograms within the full frequency band with the same parameters that were selected at low frequencies. Before performing the SI summation, we need to recover the original shape after applying the FX-interpolation. Inverting the two flattening operations is straightforward. The cross-correlogram in Figure 2c is obtained after application of our method. Now we stack the cross-correlated traces to generate one trace for large receiver interval (Figure 2d, green). For comparison, the trace reconstructed using finer receiver interval is shown in blue (Figure 2d).

In summary, our method can be described in the following steps:

1) Cross-correlating traces from a pair sources and all common receivers to generate the cross-correlogram using as first input the subsurface multiple and as second input the seafloor reflection modelling.
2) Applying a high-cut filter the cross-correlogram.
3) Selecting the quadratic flattening parameter related to wavefield spreading.
4) Selecting the linear flattening parameter related to the dipping reflectors.
5) Flattening the events in cross-correlogram with full frequency band using the selected parameters.
6) Applying the FX-interpolation.
7) Recovering the cross-correlogram shape by inverting the linear and quadratic flattening operations.
8) Applying a taper at the two ends of the cross-correlogram to alleviate for the limited-aperture artefacts.
9) Stacking the cross-correlogram to generate the virtual trace.

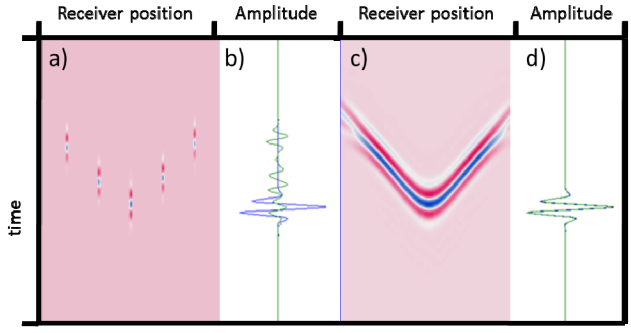


Figure 2 *(a) Cross-correlogram for traces with a large receiver spacing and in the full frequency band. (b) Green line shows trace generated by stacking the cross-correlogram in (a); Blue line shows the expected trace after stacking cross-correlation with a finer receiver spacing. (c) Cross-correlogram obtained after the FX-interpolation guided by the flattening parameters selected at lower frequencies. (d) Green line shows the trace generated by stacking (c); Blue line shows trace generated after stacking cross-correlations with a finer receiver spacing.*

Numerical examples

To demonstrate the applicability of our method, we generated a synthetic dataset for a three-layer velocity model (Figure 3a). The receivers were placed at 255 m depth at the seafloor spaced by 320 m and the shots were fired at depth of 25 m and with 20 m of interval. The source was a Ricker wavelet with the central frequency of 15 Hz.

Afterwards, we selected two traces related to one receiver and two different shots (it may be the same shot for zero-offset trace reconstruction) and cross-correlated them using as one input only the reflections of the subsurface. This input trace is obtained by subtracting the synthetic trace of the velocity model in Figure 3a and the trace generated using the flat seafloor as only reflector. Figure 3b shows a common receiver gather with receiver located at 1140 m as a result of this subtraction. The second input trace is taken from the modelling with the seafloor as only reflector. Figure 3c shows the common receiver gather at same location for flat seafloor modelling.

Those events that mainly contribute to reconstruct the traces of the shallow reflector are highlighted by arrows in Figures 3b and 3c, respectively. Then, we applied our method for different pairs of shots converting one of them into a virtual receiver. We obtained a streamer-like dataset with shot and receiver at 25 m depth and spaced every 20 m for near-offsets with a range from -320 m to 320 m.

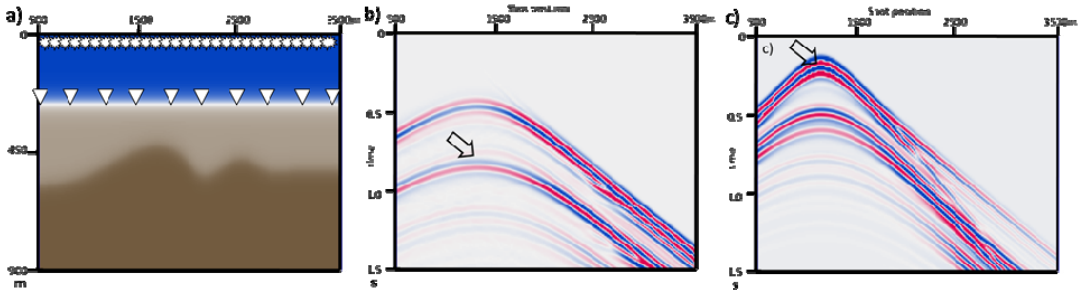


Figure 3 *(a) Zoomed-in of the velocity model used for the numerical experiment. Ocean bottom receiver spacing is 320 m, while shot interval is 20 m. (b) First input for SI, one of the common receiver gathers after subtracting the direct arrival and the SRM created by the seafloor. (c) The direct arrival and the SRM generated by the seafloor are the second input for SI. Arrows represent the SRM that mainly contribute to illuminate the reflector at 450 m in (a).*

Figure 4a shows the migrated image obtained using the dataset reconstructed with our method and receiver interval of 320 m. Performing migration on raw data with such large receiver interval will mostly create artefacts in the image due to spatial aliasing as in the reconstructed trace in Figure 2b. For comparison, we have modelled and migrated the dense streamer-like dataset with the desired geometry (Figure 4b). The structure is preserved; nonetheless, the wavelet was affected because of the cross-correlation in the migration.

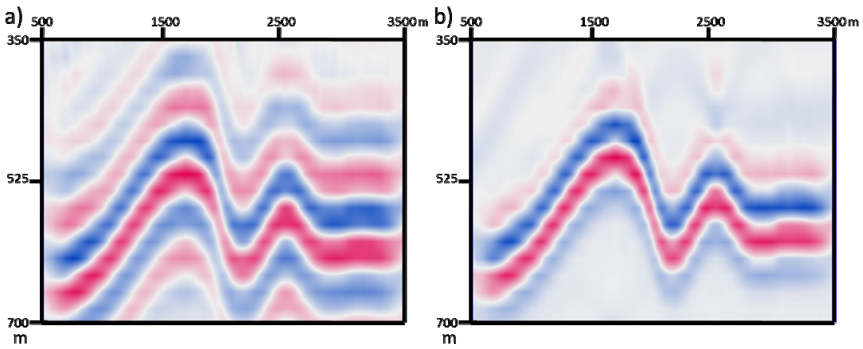


Figure 4 *(a) Migration section obtained after redatuming the multiples into primaries using the FX-interpolation and seismic interferometry for data with receiver spacing of 320 m. (b) Migration section using synthetic streamer-like data at 20 m receiver spacing.*

Conclusions

We propose using the SI to retrieve information from the SRM and to redatum them towards the primary energy. When the receiver spacing is large and the subsurface structure is complex, artefacts may appear. To mitigate this, we suggest using the FX-interpolation on the cross-correlograms before the SI summation because it provides more degrees of freedom for flattening operations before the interpolation.

With a synthetic dataset, we demonstrated that our method can redatum the multiples into primaries with a configuration of dense receiver and source arrays at the same depth. Then, the reconstructed primaries can be used to image shallow subsurface structures as dense streamer-like seismic data were acquired.

Our proposed method is data-driven. However, several hurdles still remain. A-priori results such as tomography and more robust interpolation methods can improve the application of our method on complex geological structure dataset.

Acknowledgements

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